

No. of Experiment:

Name of the Experiment: “Determination of the radius of curvature of a spherical surface with a spherometer”.

Theory:

Each spherical surface is a part of sphere whose radius is called the radius of curvature of that spherical surface. First, the spherometer is placed upon a plane glass plate and the reading is taken when screw just touches the plate. Again the spherometer is placed on the spherical convex surface and reading is taken when screw just touches the spherical surface. The height of the spherical surface $h = AC$ is obtained by taking the difference of both the readings (Fig.1). Now if the distance between the fixed leg and the screw is $XC = a$ and the radius of curvature of spherical surface is R , then it is obvious from Fig.1,

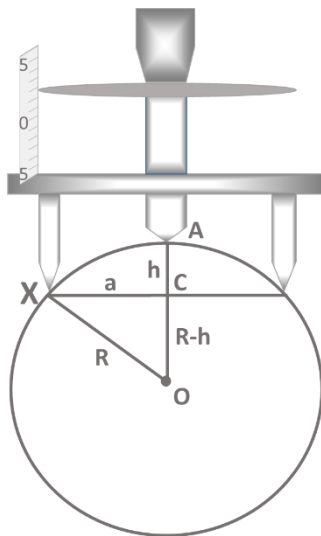


Fig. 1.

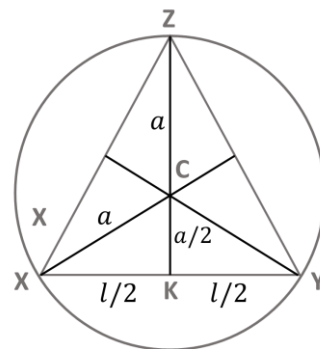


Fig. 2.

$$OX = OA = R$$

and

$$OC = OA - AC = (R - h)$$

Therefore, from the right angle triangle OCX ,

$$OX^2 = OC^2 + CX^2$$

Or

$$R^2 = (R - h)^2 + a^2$$

Or

$$2Rh = h^2 + a^2$$

$$\text{Or} \quad R = \frac{h^2}{2h} + \frac{a^2}{2h}$$

$$R = \frac{a^2}{2h} + \frac{h}{2} \quad \text{----- (i)}$$

In Fig.2, X, Y, Z represent the positions of three legs of spherometer and C is the position of its central leg (or screw). XYZ is an equilateral triangle in which the distance between two legs $XY = l$ and $CX = CY = CZ = a$. In equilateral triangle XYZ , ZK is the perpendicular bisector as well as the median both.

By the property of median, $CK = \frac{CZ}{2} = \frac{a}{2}$ and $XK = \frac{XY}{2} = \frac{l}{2}$

Now from the right angle triangle CKX ,

$$CX^2 = CK^2 + XK^2$$

$$\text{Or} \quad a^2 = \frac{a^2}{4} + \frac{l^2}{4}$$

$$\text{Or} \quad \frac{3a^2}{4} = \frac{l^2}{4}$$

$$a^2 = \frac{l^2}{3}$$

By substituting the value of a^2 in equation (i),

$$R = \frac{l^2}{6h} + \frac{h}{2}$$

By knowing l and h , R can be calculated by using the above formula.

List of Apparatus:

1. A spherometer
2. A plane glass-plate
3. A metre scale
4. A piece of white paper and
5. A given spherical surface

Procedure:

1. First, to find the pitch of the spherometer, rotated the disc by four or six complete rounds and note the distance moved by the disc on the main scale. Now divided this distance by the number of rotations to calculate the pitch.
2. Divided the pitch by the total number of divisions on the disc to calculate the least count.
3. Now placed the spherometer on a plane glass plate and rotated the central leg (or the screw) till the screw just touches the plane surface. At this position, assuming zero at the lowest mark of main scale, note the main scale reading and circular scale division which touches the edge of the main scale. After then the circular scale division were multiplied by the least count. This reading were added to the main scale reading to give the total reading.
4. Now the spherometer was placed on the given convex spherical surface such that its external three legs lie on this plane. The central leg were then turned down till it just touches the spherical surface. At this position, assuming zero at the lowest mark, note the main scale reading and the circular scale division which touches the edges of the main scale. After then the circular scale division were multiplied by the least count. This reading were then added to the main scale reading to give the total reading.
5. The height h of the spherical surface was obtained by subtracting the first reading from the second reading.

6. The spherometer was then placed upon the left page of the practical note book and is slightly pressed. Three dots corresponding to the ends of the outer fixed legs of spherometer were imprinted on the paper. By joining these three dots, an equilateral triangle was formed and by measuring the length of each side with the help of metre scale, the mean length l between the external legs of spherometer were obtained.

Experimental Data:

Calculation of least count:

The distance moved forward or backward along main scale by the screw on given one complete rotation, is known as the pitch $P = \dots\dots\dots (mm)$

Number of division on the disc $n = \dots\dots\dots$

Least count of spherometer $C = \frac{P}{n} = \dots\dots\dots (cm)$

Table 1: For the determination of height (h).

No. of Obs.	Which surface	Linear scale reading, M (cm)	Circular scale reading, V	Least count, C (cm)	Value of the circular reading, $F = V \times C$ (cm)	Total reading, $(M + F)$	Average reading, t (cm)	Height, $h = x_2 - x_1$ (cm)
1.	On the plane mirror (x_1)							
2.								
1.	On the lens (x_2)							
2.								

Determination of the distance, D between the legs of the spherometer:

$$l = \frac{l_1 + l_2 + l_3}{3} = \dots\dots\dots (cm)$$

Calculations:

$$R = \left(\frac{l^2}{6h} + \frac{h}{2} \right) = \dots\dots\dots (cm)$$

Results:

Radius of curvature of the spherical surface = (cm)

Precautions and Discussion:

1. Reading of the spherometer should be taken only when the tip of the screw just touches the plane of the plate or the spherical surface. In this position, the tip of the screw and its image on the plate will just touch each other.
2. To avoid backlash error, the screw moved in the same direction.
3. The main scale always kept in vertical.
4. For the determination of the value of l , the distance between each two consecutive of the outer three legs should be measured and their mean is calculated.